Nikita Rathour

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Summary

Full Stack Developer and AI-Engineer with hands-on experience developing agentic AI systems, RAG pipelines, and scalable web applications. Proficient in React.js, FastAPI, Python, PostgreSQL, and LangGraph/LangChain. Proven track record from an enterprise software development internship building production HRMS and LMS solutions. Seeking an entry-level Software Developer or Full Stack Engineer role.

Education

Haridwar University **India**
Bachelor of Technology in Computer Science & Engineering *2023 – 2027*

Experience

BKNetwork Services Pvt. Ltd. **Ghaziabad, India**
Software Development Intern — Python, React.js, PostgreSQL, REST APIs *June 2026 – Sept 2026*
- Engineered core modules for enterprise **HRMS** and **LMS** platforms, deploying live applications to production.
- Designed and maintained secure **RESTful APIs** using **Python** and **PostgreSQL** (DBeaver) with **JWT authentication**.
- Developed intuitive, responsive frontend interfaces with **React.js** and **Tailwind CSS** for seamless navigation.
- Collaborated with cross-functional teams in an agile environment using **Git/GitHub** for pull requests and code reviews.

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, Java, SQL, HTML5, CSS3
Frontend: React.js, Next.js, Tailwind CSS, Streamlit, Vite, Responsive Design
Backend: FastAPI, Node.js, Express.js, RESTful APIs, JWT Authentication
AI & Agents: LangGraph, LangChain, Multi-Node Agent Workflows, Hybrid RAG, ChromaDB, Gemini API
Databases & Tools: PostgreSQL, MongoDB, Git, GitHub API, DBeaver, Pytest, VS Code, Figma, Postman
Core CS: Data Structures & Algorithms, DBMS, Operating Systems, OOP

Projects

Autonomous Software Engineer Agent | *React, Python, FastAPI, LangGraph, Gemini, GitHub API* **GitHub**
- Architected an autonomous AI studio (**React**, **Vite**, **FastAPI**) that resolves GitHub Issues through automated patch synthesis.
- Engineered an 8-node **LangGraph** state machine with Human-in-the-Loop (`interrupt_before`) approval checkpoints.
- Implemented self-correcting test automation in a sandbox via **pytest**, analyzing failures and retrying synthesis up to 3 times.
- Automated Git workflows via **GitHub REST API**, generating diff patches, feature branches, and PRs with test metrics.

Multilingual Voice-Enabled Legal Assistant | *Python, LangChain, ChromaDB, Gemini, Streamlit, BM25* **GitHub**
- Engineered an end-to-end **Multilingual RAG** pipeline querying Indian statutory documents (IPC, IT Act) via voice and text.
- Implemented **Hybrid Retrieval** (**ChromaDB** dense vectors + **BM25** sparse search) with a Cross-Encoder reranker.
- Designed an **Evidence Verification & Abstention** mechanism preventing hallucinations when legal evidence is insufficient.
- Integrated **BHASHINI / TTS** speech synthesis alongside an empirical benchmark evaluating faithfulness on 110+ queries.

AI Athlete Performance & Validation System | *Python, YOLOv8, MediaPipe, OpenCV, React.js, FastAPI* **GitHub**
- Developed a real-time computer vision system using **MoveNet** and **MediaPipe** for athletic pose tracking and analysis.
- Integrated **YOLOv8** and **OpenCV** algorithms to automate cheat detection and validate biomechanical movement accuracy.
- Built a **React.js** dashboard and **FastAPI** backend with **MongoDB**, supporting offline analysis in low-bandwidth environments.

Achievements

- **ACM Shivalik Hackathon:** Secured Top 10% ranking (12th rank among 121 teams) for AI-driven Agriculture platform (ApnaKhet).
- **Smart India Hackathon (SIH):** Shortlisted for internal round for AI-Based Athlete Performance Evaluation System.